

Emotion Geometry in Embedding Spaces: Complete Experimental Documentation

1. Problem Definition

The objective of this work is to systematically analyze how emotional categories are represented in embedding spaces generated by large language models. The experiments are designed to uncover the geometric structure, separability, ambiguity behavior, dimensional dependence, and decision alignment of emotional embeddings. The study is divided into two major components: (1) LLM embedding analysis (2) Brain comparison analysis

2. Data and Representation

All experiments use a balanced dataset of 4038 samples across six emotional classes: anger, fear, happiness, love, sadness, and surprise. Each input is converted into a 768-dimensional embedding vector. All embeddings are L2-normalized before analysis to ensure consistent geometric comparisons. Mathematically: $x \in \mathbb{R}^{768} \ ||x|| = 1$

3. Experimental Pipeline (LLM Embeddings)

Step 1: Embedding Extraction

Embeddings are extracted from transformer models at both intermediate and final layers. Two categories are used: - Pretrained models (no task-specific supervision) - Fine-tuned models (trained on emotion labels) Each sample is passed through the model to obtain a vector representation.

Step 2: Centroid Computation

For each class, a centroid is computed as the mean of all embeddings belonging to that class: $c_k = (1/N_k) * \sum x_i$ where k is the class index.

Step 3: Distance Computation

Distances between points and centroids are computed using Euclidean distance. These distances are used for clustering evaluation and overlap analysis.

4. Phase 1: Baseline Topology

Goal: Measure how well emotional categories are separated in embedding space. Metric: Silhouette Score Key Findings: - Fine-tuned models: strong cluster separation (≈ 0.85) - Pretrained models: weak separation (≈ 0.05) Interpretation: Fine-tuning creates discrete clusters, while pretraining results in overlapping semantic regions.

5. Phase 2: Overlap and Ambiguity Gradient

Goal: Measure confusion between emotional classes. Step-by-step: 1. Compute distance to true centroid 2. Compute distance to all other centroids 3. Count cases where incorrect centroid is closer Metrics: - Overlap % - Radial density distribution - Ambiguity gradient Findings: - Pretrained: 18–28% overlap - Fine-tuned: 1–2% overlap Structure observed: - Core region: no ambiguity - Transition region: gradual confusion - Outer region: high ambiguity

6. Phase 3: Signal Retention Analysis

Goal: Identify how emotional information is distributed across dimensions. Steps: 1. Train linear classifier 2. Perform SVD on weights 3. Remove top singular directions 4. Recompute accuracy Findings: - Fine-tuned models collapse after ~15–20 dimensions - Pretrained models retain signal up to ~60 dimensions Interpretation: Fine-tuned models compress information; pretrained models distribute it.

7. Phase 4: Low-Dimensional Subspace Analysis

Goal: Test if emotional information lies in a smaller subspace. Steps: 1. Extract top 20 singular vectors 2. Project embeddings 3. Recompute clustering and accuracy Findings: - Huge improvement for pretrained models (+600%) - Minimal loss of accuracy (~95–98%) Conclusion: Emotion is encoded in a low-dimensional manifold.

8. Phase 5: Geometry vs Decision Alignment

Goal: Determine if model predictions follow geometric structure. Steps: 1. Identify overlapping samples 2. Compare logits with distances 3. Compute correlation Findings: - Correlation: 0.95–0.98 - Agreement: up to 100% Conclusion: Model decisions are strongly governed by geometry.

9. Brain Comparison Experiments

The same analyses are conceptually applied to brain representations. Important terminology: - Fine-tuned model = supervised representation - Pretrained model = unsupervised representation Comparisons performed: 1. Geometry similarity 2. Ambiguity gradient comparison 3. Centroid distance alignment 4. Subspace similarity 5. Valence-arousal correlation Findings: - Brain behavior aligns more

closely with pretrained models - Emotional representation is distributed, not strictly categorical - Ambiguity structure is preserved

10. Final Conclusions

1. Emotional representations are inherently geometric. 2. Fine-tuning sharpens but compresses representations. 3. Pretrained systems maintain robustness and redundancy. 4. Emotional information lies in a low-dimensional structure. 5. Geometry governs both representation and decision-making.